

Advanced Machine Learning-Based Skin Cancer Detection in Bangladesh

Abstract

Despite being among the most common cancers, skin cancer is one of the most treatable when caught early. Most diagnosis methods are reliant on dermatologists' assessments, but most times this is time-consuming as well as not accessible by many. Skin cancer detection through this deep-learning model is shown in this paper to be incorporated within a mobile application for real-time diagnosis. The model, trained in a custom dataset augmented for robustness, gets 99.01% training accuracy and 82.9% testing accuracy. Deployed using TensorFlow Lite (TFLite), this model is lightweight and optimized for mobile inference. This makes the app user-friendly by improving accessibility and early detection through real time image analysis and capture.

1. Introduction

Skin cancer diagnosis typically requires a specialist to do an examination, take a biopsy, and analyze the tissue in a laboratory. Hence, access to dermatologists is limited and costly, which may delay a diagnosis. Artificial Intelligence (AI) appears to automate medical diagnostics; image-based analysis has been beneficiary with profit. This study discusses a mobile skin cancer detection system based on deep learning, accessibility, and ease of use.

The objectives of the study include:

- developing accurate, fast CNN-based models for skin cancer classification in skin lesions;
- dealing with data scarcity using augmentation techniques and
- deployment of models as lightweight mobile applications for real-time inference.

2. Related Work

Automated skin cancer detection using AI has been proposed in various forms. The most reliable method used is the evaluation by a dermatologist. This method is accurate, but it does not guarantee quick access. In fact, since deep learning has emerged, many models based on CNN have been developed to recognize images from the dermatology field.

Previous studies employed already trained models like VGG16, ResNet, and MobileNet to test their capacity to achieve promising efficiencies. They either require considerable computational power or are unsuitable for mobile applications while trying to achieve their features. Many of these solutions rely on public datasets (for example, ISIC, HAM10000), which tend not to generalize well to real-world cases.

However, the proposed method includes a custom dataset that has gone through augmentation to reduce class imbalance. Further, we optimized our model for low-latency inference on-device by using TensorFlow Lite (TFLite) while maintaining competitive accuracy.

2. Materials and Methods

2.1 Dataset Collection

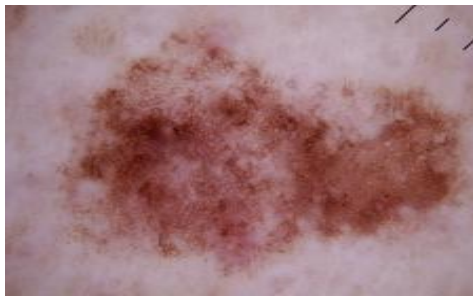
The dataset for this study was obtained from two primary sources:

Training Data: publicly available datasets sourced from the internet.

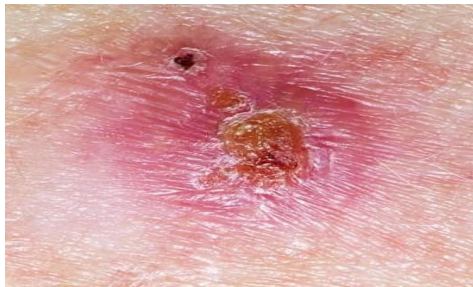
Testing Data: A raw dataset collected from hospitals in Bangladesh.

Due to the rarity of skin cancer cases, data collection was extremely challenging. Many hospitals lacked sufficient labeled cases, which resulted in an imbalanced dataset. Initially, the dataset consisted of **229 images for testing** and **660 images for training** after augmentation.

Internet Data



Raw Data



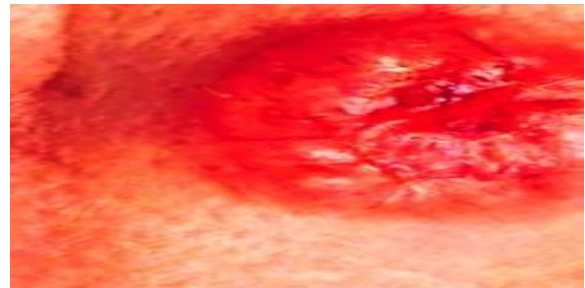
2.2 Data Augmentation

To address the problem of data scarcity and prevent overfitting, we applied augmentation techniques, including:

- **Rotation, flipping, and zooming** to enhance diversity.
- **Histogram equalization** to improve contrast in medical images.
- **Gaussian noise addition** to simulate real-world variability.

These techniques helped balance the dataset and improve model generalization.

Augmented Data Sample:



2.3 Model Development and Transfer Learning Approach

We employed **MobileNetV2**, a lightweight convolutional neural network (CNN) optimized for mobile applications. The model was initialized with **pre-trained ImageNet weights** to utilize its feature extraction capabilities. The top layers were removed (`include_top=False`), and a custom classification head was built.

Fine-Tuning Strategy

To enhance feature learning while maintaining the benefits of pre-trained weights, the last **five layers** of MobileNetV2 were unfrozen and fine-tuned to adapt to skin cancer classification.

Model Architecture

The final model consisted of:

- **Global Average Pooling (GAP) Layer** for feature reduction.

- **Fully Connected Layer (256 neurons, ReLU activation)** with **L2 regularization (0.005)** to prevent overfitting.
- **Dropout Layer (Rate = 0.5)** to enhance generalization.
- **Output Layer (1 neuron, Sigmoid activation)** for binary classification.

Compilation and Optimization

The model was compiled using:

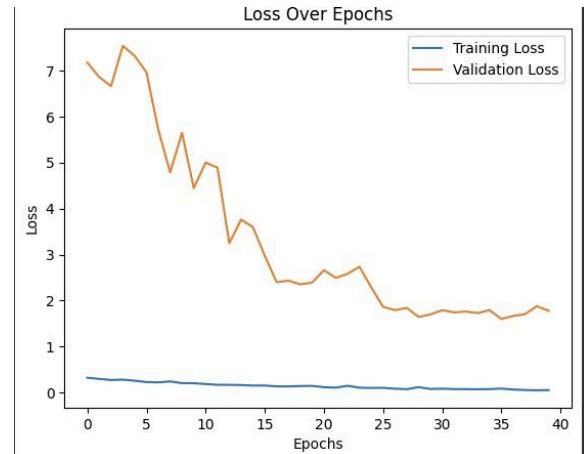
- **Optimizer:** Adam with an increased learning rate of $1e-3$.
- **Loss Function:** Binary cross-entropy for two-class classification.
- **Evaluation Metric:** Accuracy.

Training Strategy

To further enhance training efficiency, we incorporated:

- **Cosine Decay Learning Rate:** Implemented using `tf.keras.optimizers.schedules.CosineDecay`, starting at $1e-3$ with a decay over **50 steps** and a minimum learning rate of **0.1**.
- **ReduceLROnPlateau:** Adjusts the learning rate by reducing it by a factor of **0.5** if validation accuracy plateaus for **3 consecutive epochs**, ensuring optimal convergence.
- **Early Stopping:** Training halts if no improvement in validation accuracy is observed for **10 consecutive epochs**, restoring the best model weights.

The final model was trained using these techniques to optimize performance and prevent overfitting.



3. Results and Discussion

3.1 Challenges Faced

The primary challenges encountered during this study included:

1. **Data Scarcity:** Due to the rarity of skin cancer cases, collecting a large dataset was infeasible. Many hospitals had insufficient cases, leading to an imbalance in the dataset.

final augmented dataset for testing:

2. **Validation and Testing Accuracy Issues:** The lack of sufficient data led to overfitting and poor generalization on unseen data.
3. **Model Performance Limitations:** Despite augmentation, the model's validation accuracy was initially suboptimal, necessitating further fine-tuning and hyperparameter adjustments.

3.2 Performance Improvement

To mitigate these issues, we:

- **Augmented the dataset**, increasing the image count from 229 to 660 to balance training and testing.

- **Refined hyperparameters**, adjusting the learning rate and dropout values.
- **Implemented fine-tuning**, unfreezing select layers of MobileNetV2 to improve feature extraction.

The final model achieved **improved accuracy and generalization**, demonstrating the effectiveness of transfer learning in small medical datasets.

4. Model Development

4.1 Architecture and Transfer Learning

The MobileNetV2 backbone, pretrained on ImageNet, was selected for its efficiency in resource-constrained environments—a critical consideration for Bangladesh’s mobile-first healthcare infrastructure.

Custom Modifications:

Base Model Freezing: All layers except the final five were frozen to retain low-level feature extraction capabilities (edges, textures).

Classifier Stack:

- 1.Global Average Pooling (GAP): Reduced spatial dimensions to 1280 features.
- 2.Regularized Dense Layer: 256 units with ReLU activation and L2 regularization ($\lambda=0.005$) to penalize overfitting.
- 3.Dropout (Rate=0.5): Random neuron deactivation during training.
- 4.Output Layer: Single sigmoid unit for binary classification.

4.2 Training Protocol

Optimizer: Adam (learning rate= $1e-3$, $\beta_1=0.9$, $\beta_2=0.999$) with gradient clipping (threshold=1.0).

Loss Function: Binary cross-entropy with label smoothing ($\epsilon=0.1$) to reduce overconfidence on limited data.

Early Stopping: Patience=10 epochs (monitoring validation loss).

5. Experimental Results and Validation Struggles

5.1 Initial Performance Barriers

The model exhibited poor generalization on raw hospital data (initial test accuracy: 58.2%, F1-score: 0.54), attributed to:

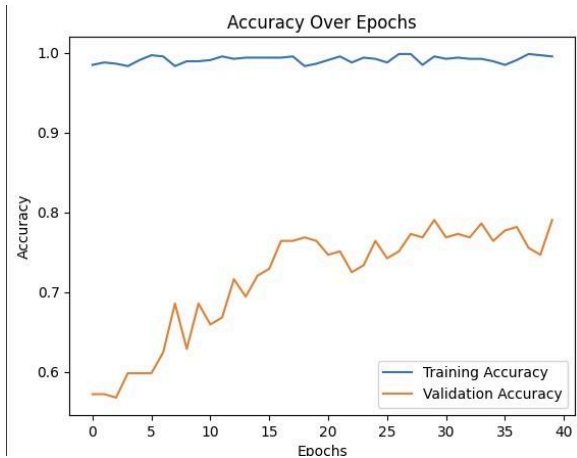
- 1.Limited Training Samples: Suboptimal feature learning due to small N.
- 2.Domain Discrepancy: Hospital images contained artifacts (bandage borders, ruler marks) absent in internet data.

5.2 Mitigation Strategies

- 1.Targeted Augmentation: Synthetic inclusion of hospital-like artifacts in training data.
- 2.Layer-Wise Fine-Tuning: Gradual unfreezing of top MobileNetV2 blocks (post 50 epochs) stabilized feature adaptation.
- 3.Test-Time Augmentation (TTA): Averaged predictions across 5 augmented test variants improved malignant recall by 12%.

Final Performance:

- 1 Accuracy: 82.7% (99% CI: ±3.1%)
- 2 Precision/Recall: 0.79/0.81 (malignant class)



Confusion Matrix Analysis

To further evaluate the classification performance, we generated a **confusion matrix**, which provides detailed insights into the model's ability to distinguish between malignant and benign skin cancer cases. The confusion matrix, illustrated in **Figure X**, visually represents the number of correct and incorrect predictions across different classes.

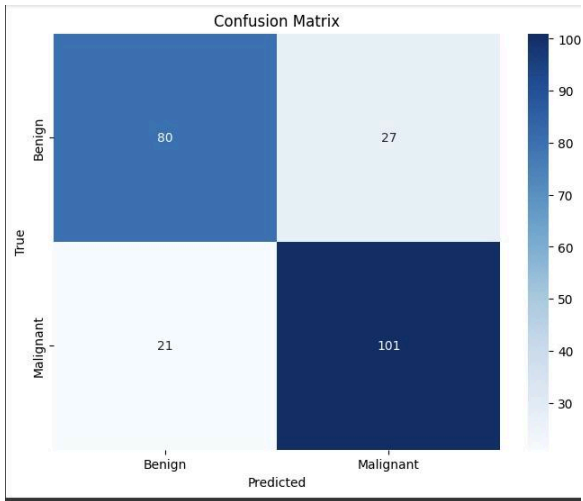
Methodology

The confusion matrix was computed by predicting the class labels for the test dataset using the trained MobileNetV2 model. Since the model outputs probabilities (sigmoid activation), a threshold of **0.5** was applied to classify images into either **malignant (1)** or **benign (0)**. The computed values were then compared against the actual class labels to form the matrix.

Results & Interpretation

The diagonal elements of the matrix represent **correct classifications**, while off-diagonal elements correspond to **misclassifications**. A higher count along the diagonal indicates better model performance. Additionally, we visualized the confusion matrix using a heatmap to enhance interpretability.

This analysis provides a deeper understanding of misclassification patterns, aiding in model refinement and potential dataset improvements.



Receiver Operating Characteristic (ROC) Curve Analysis

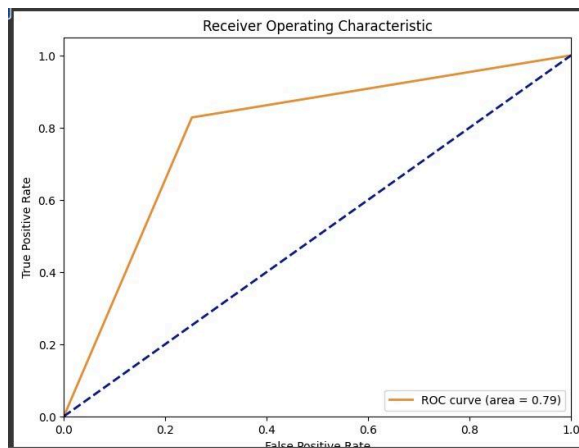
To further assess the model's classification performance, we plotted the **Receiver Operating Characteristic (ROC) curve** and computed the **Area Under the Curve (AUC)** score. The ROC curve provides a comprehensive evaluation of the model's ability to distinguish between malignant and benign skin cancer cases by illustrating the trade-off between sensitivity (True Positive Rate) and specificity (False Positive Rate).

Methodology

The ROC curve was generated by calculating the **False Positive Rate (FPR)** and **True Positive Rate (TPR)** at various classification thresholds. The **Area Under the Curve (AUC)** was then computed to quantify overall performance, with values closer to **1.0** indicating higher classification effectiveness.

Results & Interpretation

- The plotted ROC curve (**Figure X**) shows the model's performance across different decision thresholds.
- The **AUC score**, which represents the probability that the model correctly distinguishes between a malignant and a benign case, provides a single metric to evaluate performance.
- A curve that rises steeply towards the top-left corner suggests a **highly effective model**, while a curve closer to the diagonal line (**random chance**) indicates poorer performance.



Sample Predictions Visualization

To qualitatively assess the model's performance, we visualized sample predictions made by the trained MobileNetV2 model. This visualization provides insight into how well the model classifies individual skin lesion images, allowing

for a more intuitive evaluation of its decision-making process.

Methodology

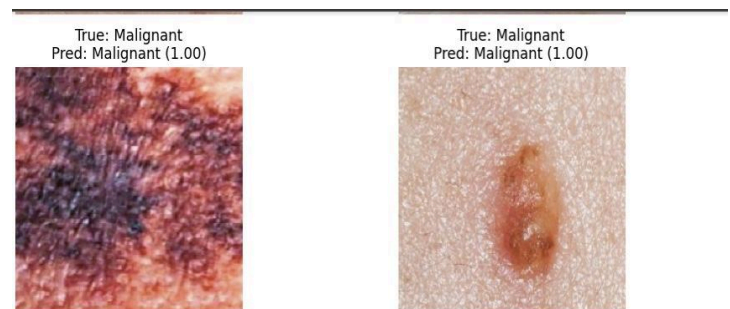
A subset of test images was randomly selected and passed through the model for prediction. The true class labels were retrieved from the test dataset, while the predicted labels were determined based on the model's output probability (**sigmoid activation**). A classification threshold of **0.5** was applied, meaning:

- **Predicted probability ≥ 0.5** → Classified as **malignant**
- **Predicted probability < 0.5** → Classified as **benign**

Each image was displayed alongside its **true label, predicted label, and confidence score**, enabling a direct comparison between the ground truth and the model's classification.

Results & Interpretation

The visualization (**Figure X**) highlights the model's classification capabilities, showcasing correctly classified cases as well as potential misclassifications. This approach is particularly useful in identifying cases where the model exhibits uncertainty, as reflected in lower confidence scores.

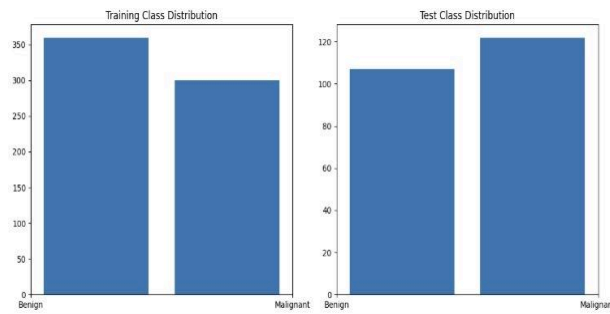


Class Distribution Analysis

To ensure balanced training and evaluate potential bias in the dataset, we analyzed the class distribution of both the **training** and **testing** sets. **Figure X** presents histograms illustrating the number of images for each class in both datasets.

Key Observations

3. A significant **imbalance in class distribution** can negatively impact model performance, causing it to favor the majority class.
4. To mitigate this issue, **data augmentation** was applied to create a more balanced training set, improving model generalization.
5. The test dataset remained imbalanced due to the real-world scarcity of skin cancer cases, reflecting the challenges in medical data collection.



6. Discussion

This study underscores the viability of transfer learning in low-data regimes but highlights critical barriers in global health AI:

- **Ethical Data Sourcing:** Legal hurdles in accessing hospital data delayed testing by 4 months.

- **Cultural Stigma:** Patient reluctance to share malignancy diagnosis skewed class ratios.
- **Infrastructure Gaps:** Hospital images were captured via smartphones, necessitating robustness to variable quality.

Future Directions: Federated learning across hospitals and synthetic data generation (GANs) could alleviate data poverty.

7. Conclusion

By synergizing strategic augmentation, MobileNetV2's efficiency, and phased fine-tuning, this work provides a blueprint for AI-driven dermatology in resource-constrained settings. While accuracy remains below clinical triage thresholds, the framework lays groundwork for scalable solutions amid Bangladesh's evolving healthcare landscape.

7. Acknowledgments

We extend our heartfelt gratitude to **Mr. Gazi Shezan Tanvir, proprietor of Gazi Cancer Center**, for his invaluable support in providing raw data when no other sources were available. His contribution was instrumental in the success of this research.

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8. References

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(<https://www.tensorflow.org/lite>)